Gathering execution technical metrics in ELIXIR-STEERS: from low-level metrics to measuring software usage's carbon footprint

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(this poster)

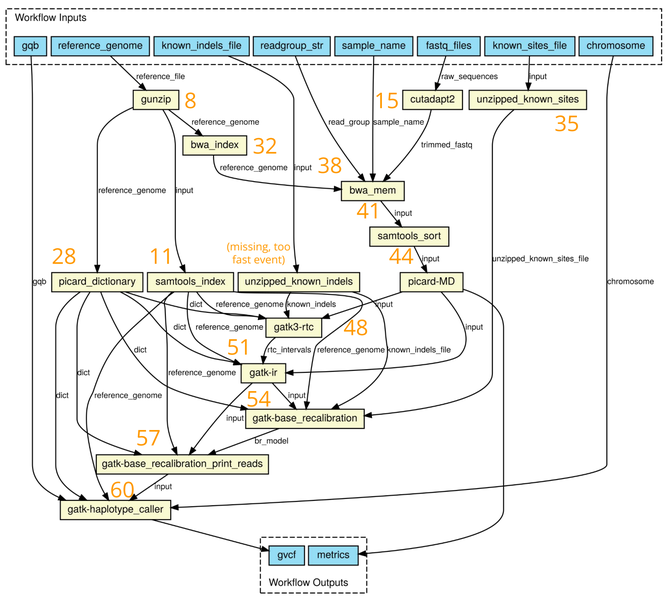
How can we learn about the impact of different components from complex analytical pipelines?

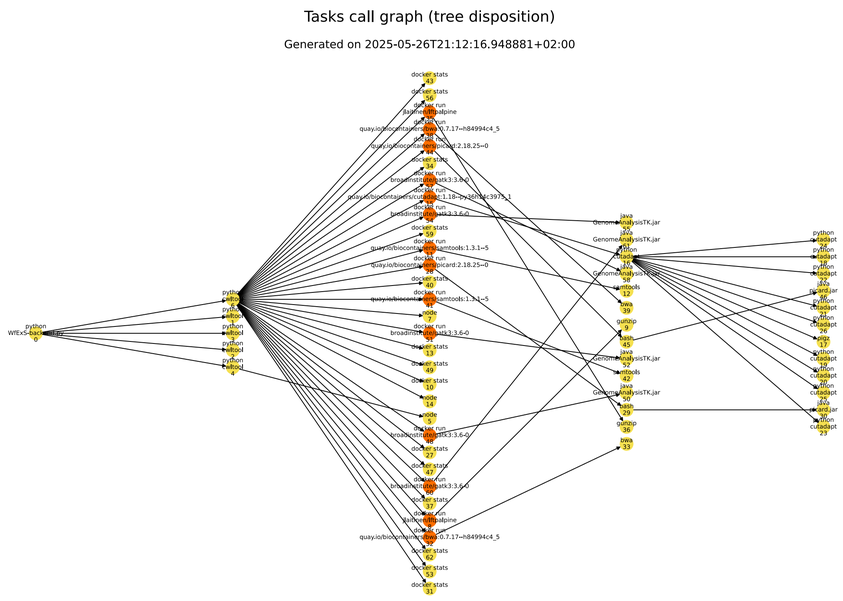
Profiling a complex set of software pieces can be complicated, even in a single computer. The approach we have followed at ELIXIR STEERS 3.3 is gathering metrics from bottom to top, at the level of every process being involved in the analysis.

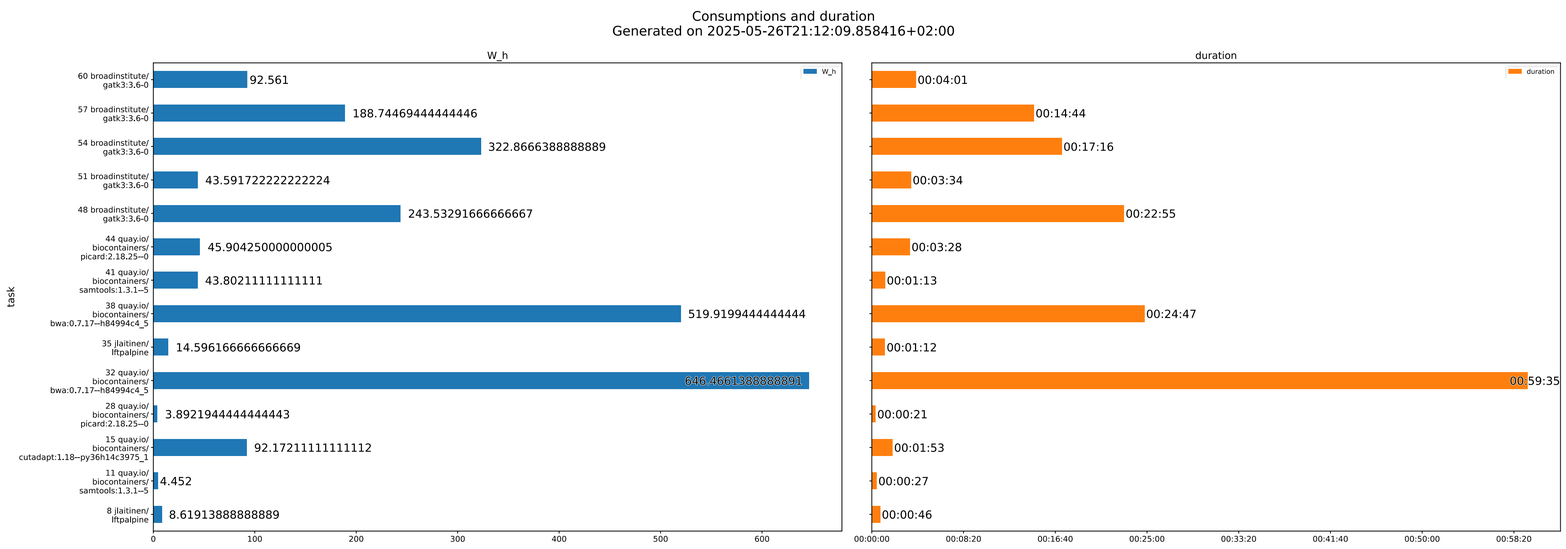
- Data series of low level metrics gathered every second: memory (virtual and resident) usage, CPU usage, task placement, I/O, ...
- Details about the CPUs and cores where the analysis was run.
- Relationships among the different processes.

execution-process-metrics-collector

Example of scientific workflow, execution call graph and gathered consumptions







Heuristics to identify different computational units

A. **Container-Based Detection:** If a workflow relies on a containerization tool (e.g., Docker, Apptainer, Singularity, Podman), identify relevant commands in the process tree timeline and compute aggregated metrics for each corresponding process subtree.

B. **Depth-Based Heuristic:** Use domain knowledge to select subtrees of processes starting at a specific depth in the process tree timeline as computational units.

C. **Parent-Child Heuristic:** Treat the subtree rooted at a specific process (and including all its descendants) as a distinct computational unit.

Low and high level metrics

- **Low level metrics series**
  Time, PID, Virtual Memory Size, Resident Set Size, CPU, Memory, TCP connections, Thread Count, User, System, Children_User, Children_System, IO, Unique Set Size, swap, cpu_num

- **High level metrics** - estimating energy consumption and CO_2 emissions.
  Requirements:
  - Power Usage Effectiveness (PUE) of the Facility
  - Average Carbon Intensity
  - Cost and Currency of Electricity (kWh)
  - Type of Processing Core:
    - Processing Core TDP (Thermal Design Power)
    - Processing Core Model
  - Granularity of Memory Request

Aggregated metrics from a subprocess tree (proposed)

- Peak and average parallelism level.
- Average and peak memory consumptions.
- Percentage of I/O time vs CPU time vs idle time.
- Estimated consumed energy and carbon footprint.

... and the ones you can think for your analyses!

Community proposed use cases

Workflow / Engine	ELIXIR Community
nf-core/scrnaseq	Single Cell Community
Scipion	3DBioinfo Community
CAID	IDP Community
Wombat-P workflows	Proteomics Community
MGnify amplicon analysis pipeline v6.0	Microbiome Community

Metrics from several, similar subprocess trees from different parameters

- *Different inputs, same parameters*: metrics evolution based either on input complexity or file size or number of input files.
- *Same input, different parameters*: metrics evolution based on number of allowed parallel processors, on max memory allowed, etc...
- *Same input, different pipelines with similar functionality*: both technical and scientific metrics (your imagination puts the limits!).

Contact

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